

Faculdade de Ciências - Departamento de Computação

Disciplina: Pesquisa Operacional

Docente: Douglas Rodrigues

Lista de Exercícios - L6

Método de Gomory

Instruções:

- A lista é individual.
- A resolução deve ser feita de próprio punho (manuscrita).
- A lista deve ser entregue no formato PDF.
- O arquivo deve ser renomeado com o nome e sobrenome seguidos pela letra “L” e o número da lista.
- Deve-se utilizar letras minúsculas separados por traço baixo (underline).
- **Exemplo:** douglas_rodrigues_l6.pdf

Exercícios

1. Utilize o método de Gomory para resolver os seguintes problemas:

(a)

$$\begin{aligned} \max z &= 2x_1 + 10x_2 + x_3 \\ \text{sujeito a: } &\begin{cases} 5x_1 + 2x_2 + 1x_3 \leq 15 \\ 2x_1 + 1x_2 + 7x_3 \leq 20 \\ 1x_1 + 3x_2 + 2x_3 \leq 25 \\ \vec{x} \in \mathbb{Z}_+^3 \end{cases} \end{aligned}$$

(b)

$$\begin{aligned} \max z &= 1x_1 + 9x_2 + x_3 \\ \text{sujeito a: } &\begin{cases} x_1 + 2x_2 + 3x_3 \leq 9 \\ 3x_1 + 2x_2 + 2x_3 \leq 15 \\ \vec{x} \in \mathbb{Z}_+^3 \end{cases} \end{aligned}$$

(c)

$$\begin{aligned} \max z &= 7x_1 + 7x_2 + 6x_3 + 9x_4 \\ \text{sujeito a: } &\begin{cases} 4x_1 + 5x_2 + 3x_3 + 5x_4 \leq 300 \\ 2x_1 + \frac{3}{2}x_2 + 3x_3 + 3x_4 \leq 200 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases} \end{aligned}$$

(d)

$$\begin{aligned} \max z &= 3x_1 + 4x_2 \\ \text{sujeito a: } &\begin{cases} 2x_1 + 1x_2 \leq 6 \\ 2x_1 + 3x_2 \leq 9 \\ \vec{x} \in \mathbb{Z}_+^2 \end{cases} \end{aligned}$$

Lista de Exercícios 6 - Método de Gomory

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1-

a) $\max z = 2x_1 + 10x_2 + x_3$

$$\text{s.a.: } \begin{cases} 5x_1 + 2x_2 + 1x_3 \leq 15 \\ 2x_1 + 1x_2 + 7x_3 \leq 20 \\ 1x_1 + 3x_2 + 2x_3 \leq 25 \\ \vec{x} \in \mathbb{Z}_+^3 \end{cases}$$

$\min z' = -2x_1 - 10x_2 - x_3$

$$\text{s.a.: } \begin{cases} 5x_1 + 2x_2 + 1x_3 + x_4 = 15 \\ 2x_1 + 1x_2 + 7x_3 + x_5 = 20 \\ 1x_1 + 3x_2 + 2x_3 + x_6 = 25 \\ \vec{x} \in \mathbb{Z}_+^3 \end{cases}$$

	x_1	x_2	x_3	x_4	x_5	x_6	b	
								$\min(15/2, 20, 25/3)$
x_4	5	2	1	1	0	0	15	$L_1 = L_1 \div 2$
x_5	2	1	7	0	1	0	20	$L_2 = L_2 - L_1'$
x_6	1	3	2	0	0	1	25	$L_3 = L_3 - 3L_1'$
	-2	-10	-1	0	0	0	0	$L_4 = L_4 + 10L_1'$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_2	$5/2$	1	$1/2$	$1/2$	0	0	$15/2$
x_5	$-1/2$	0	$13/2$	$-1/2$	1	0	$25/2$
x_6	$-13/2$	0	$1/2$	$-3/2$	0	1	$5/2$
	23	0	4	5	0	0	75

$$5/2 x_1 + 1/2 x_3 + 1/2 x_4 = 15/2$$

$$(2 + 1/2)x_1 + (0 + 1/2 x_5) + (0 + 1/2)x_4 = (7 + 1/2)$$

$$1/2 x_1 + 1/2 x_3 + 1/2 x_4 \geq 1/2$$

$$x_1 + x_3 + x_4 - x_7 + w_1 = 1$$

$$\min z^a = w_1 = -x_1 - x_3 - x_4 + x_7 = 1$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b	$\min(3, 1)$
x_2	$5/2$	1	$1/2$	$1/2$	0	0	0	0	$15/2$	$L_1 = L_1 - 5/2 L_4$
x_5	$-1/2$	0	$13/2$	$-1/2$	1	0	0	0	$25/2$	$L_2 = L_2 + 1/2 L_4$
x_6	$-13/2$	0	$1/2$	$-3/2$	0	1	0	0	$5/2$	$L_3 = L_3 + 13/2 L_4$
$\rightarrow w_1$	1	0	1	1	0	0	-1	1	1	$L_4 = L_4$
z^a	-1	0	-1	-1	0	0	1	0	z^{a-1}	$L_5 = L_5 + L_4$
z'	23	0	4	5	0	0	0	0	75	$L_6 = L_6 - 23 L_4$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b
x_2	0	1	-2	-2	0	0	$5/2$	$-5/2$	5
x_5	0	0	7	0	1	0	$-1/2$	$1/2$	13
x_6	0	0	7	5	0	1	$-13/2$	$13/2$	9
x_1	1	0	1	1	0	0	-1	1	1
z^a	0	0	0	0	0	0	0	1	z^a
z'	0	0	-19	-18	0	0	23	-23	520

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b	$\min(13/7, 9/7, 1)$
x_2	0	1	-2	-2	0	0	$5/2$	5	$L_1 = L_1 + 2 L_4$
x_5	0	0	7	0	1	0	$-1/2$	13	$L_2 = L_2 - 7 L_4$
x_6	0	0	7	5	0	1	$-13/2$	9	$L_3 = L_3 - 7 L_4$
$\rightarrow x_1$	1	0	1	1	0	0	-1	1	$L_4 = L_4$
z'	0	0	-19	-18	0	0	23	520	$L_5 = L_5 + 19 L_4$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_2	2	1	0	0	0	0	$1/2$	7
x_5	-7	0	0	-7	1	0	$13/2$	6
x_6	-7	0	0	-2	0	1	$1/2$	2
x_3	1	0	1	1	0	0	-1	1
z'	19	0	0	1	0	0	4	71

$$x^* = (0 \ 7 \ 1 \ 0 \ 6 \ 2 \ 0) \quad \text{e} \quad z^* = 71 //$$

b) $\max z = 1x_1 + 9x_2 + x_3$

s.a.: $\begin{cases} x_1 + 2x_2 + 3x_3 \leq 9 \\ 3x_1 + 2x_2 + 2x_3 \leq 15 \\ \vec{x} \in \mathbb{Z}_+^3 \end{cases}$

$\min z' = -x_1 - 9x_2 - x_3$

s.a.: $\begin{cases} x_1 + 2x_2 + 3x_3 + x_4 = 9 \\ 3x_1 + 2x_2 + 2x_3 + x_5 = 15 \\ \vec{x} \in \mathbb{Z}_+^5 \end{cases}$

	x_1	x_2	x_3	x_4	x_5	b	
$\rightarrow x_4$	1	2	3	1	0	9	$L_1 = L_1 \div 2$
x_5	3	2	2	0	1	15	$L_2 = L_2 - 2L_1$
	-1	-9	-1	0	0	0	$L_3 = L_2 + 9L_1$

$\min (9/2, 15/2)$

	x_1	x_2	x_3	x_4	x_5	b
x_2	$1/2$	1	$3/2$	$1/2$	0	$9/2$
x_5	2	0	-1	-1	1	6
	$7/2$	0	$25/2$	$9/2$	0	$81/2$

$x_2: 1/2 x_1 + 3/2 x_3 + 1/2 x_4 = 9/2$

$(0 + 1/2)x_1 + (1 + 1/2)x_3 + (0 + 1/2)x_4 = (4 + 1/2)$

$1/2 x_1 + 1/2 x_3 + 1/2 x_4 \geq 1/2 \rightarrow x_1 + x_3 + x_4 - x_6 + w_1 = 1$

$\min z^a = w_1 = -x_1 - x_3 - x_4 + x_6 = 1$

	x_1	x_2	x_3	x_4	x_5	x_6	w_1	b	
x_2	$1/2$	1	$3/2$	$1/2$	0	0	0	$9/2$	$L_1 = L_1 - 1/2 L_3$
x_5	2	0	-1	-1	1	0	0	6	$L_2 = L_2 - 2L_3$
$\rightarrow w_1$	1	0	1	1	0	-1	1	1	$L_3 = L_3$
z^a	-1	0	-1	-1	0	1	0	$z^a - 1$	$L_4 = L_4 + L_3$
z'	$7/2$	0	$25/2$	$9/2$	0	0	0	$81/2$	$L_5 = L_5 - 7/2 L_3$

$\min (9, 3, 1)$

	x_1	x_2	x_3	x_4	x_5	x_6	w_i	b
x_2	0	1	1	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	4
x_5	0	0	-3	-3	1	2	-2	4
x_1	1	0	1	1	0	-1	1	1
z^*	0	0	0	0	0	0	1	z^*
z'	0	0	9	1	0	$\frac{7}{2}$	$-\frac{7}{2}$	37

$$-\begin{matrix} \frac{7}{2} & 0 & \frac{25}{2} & \frac{9}{2} & 0 & 0 & 0 & \frac{8}{2} \\ \frac{7}{2} & 0 & \frac{7}{2} & \frac{7}{2} & 0 & -\frac{7}{2} & \frac{7}{2} & \frac{7}{2} \end{matrix}$$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_2	0	1	1	0	0	$\frac{1}{2}$	4
x_5	0	0	-3	-3	1	2	4
x_1	1	0	1	1	0	-1	1
z'	0	0	9	1	0	$\frac{7}{2}$	37

$$x^* = (1 \ 4 \ 0 \ 0 \ 4 \ 0) \quad e \quad z^* = 37 //$$

c) $\max z = 7x_1 + 7x_2 + 6x_3 + 9x_4$
s.a.: $\begin{cases} 4x_1 + 5x_2 + 3x_3 + 5x_4 \leq 300 \\ 2x_1 + \frac{3}{2}x_2 + 3x_3 + 3x_4 \leq 200 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases}$

$\min z' = -7x_1 - 7x_2 - 6x_3 - 9x_4$
s.a.: $\begin{cases} 4x_1 + 5x_2 + 3x_3 + 5x_4 + x_5 = 300 \\ 2x_1 + \frac{3}{2}x_2 + 3x_3 + 3x_4 + x_6 = 200 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases}$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_5	4	5	3	5	1	0	300
x_6	2	$\frac{3}{2}$	3	3	0	1	200
	-7	-7	-6	-9	0	0	0

$$\min(60, \frac{200}{3})$$

$$l_1 = L_1 \div 5$$

$$l_2 = L_1 - 3l_1'$$

$$l_3 = L_3 + 9l_1'$$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	$\frac{4}{5}$	1	$\frac{3}{5}$	1	$\frac{1}{5}$	0	60
x_6	$-\frac{2}{5}$	$-\frac{3}{2}$	$\frac{6}{5}$	0	$-\frac{3}{5}$	1	20
	$\frac{1}{5}$	2	$-\frac{3}{5}$	0	$\frac{9}{5}$	0	540

$$\min(100, \frac{50}{3})$$

$$l_1 = L_1 - \frac{3}{5}L_2'$$

$$l_2 = L_2 \cdot \frac{5}{6}$$

$$l_3 = L_3 + \frac{3}{5}L_2'$$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_4	1	$\frac{7}{4}$	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	50
x_3	$-\frac{1}{3}$	$-\frac{5}{4}$	1	0	$-\frac{1}{2}$	$\frac{5}{6}$	$\frac{50}{3}$
	0	$\frac{5}{4}$	0	0	$\frac{3}{2}$	$\frac{1}{2}$	550

$$x_3: -\frac{1}{3}x_1 - \frac{5}{4}x_2 - \frac{1}{2}x_5 + \frac{5}{6}x_6 = \frac{50}{3}$$

$$(-1 + \frac{2}{3})x_1 + (-2 + \frac{3}{4})x_2 + (-1 + \frac{1}{2})x_5 + (0 + \frac{5}{6})x_6 = (16 + \frac{2}{3})$$

$$\frac{2}{3}x_1 + \frac{3}{4}x_2 + \frac{1}{2}x_5 + \frac{5}{6}x_6 \geq \frac{2}{3}$$

$$\frac{2}{3}x_1 + \frac{3}{4}x_2 + \frac{1}{2}x_5 + \frac{5}{6}x_6 - x_7 + w_1 = \frac{2}{3}$$

$$\min z^a = w_1 = \frac{2}{3} - \frac{2}{3}x_1 - \frac{3}{4}x_2 - \frac{1}{2}x_5 - \frac{5}{6}x_6 + x_7$$

↓

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b	
x_4	1	$\frac{7}{4}$	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	50	$L_1 = L_1 + \frac{1}{2}L_3'$
x_3	$-\frac{1}{3}$	$-\frac{5}{4}$	1	0	$-\frac{1}{2}$	$\frac{5}{6}$	0	0	$\frac{50}{3}$	$L_2 = L_2 - \frac{5}{6}L_3'$
→ w_1	$\frac{2}{3}$	$\frac{3}{4}$	0	0	$\frac{1}{2}$	$\frac{5}{6}$	-1	1	$\frac{2}{3}$	$L_3 = L_3 \cdot \frac{6}{5}$
z^a	$-\frac{2}{3}$	$-\frac{3}{4}$	0	0	$-\frac{1}{2}$	$-\frac{5}{6}$	1	0	$z^a - \frac{2}{3}$	$L_4 = L_4 + \frac{5}{6}L_3'$
z^1	0	$\frac{5}{4}$	0	0	$\frac{3}{2}$	$\frac{1}{2}$	0	0	550	$L_5 = L_5 - \frac{1}{2}L_3'$

$\min (20, \frac{4}{5})$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_1	b
x_4	$\frac{7}{5}$	$\frac{11}{5}$	0	1	$\frac{4}{5}$	0	$-\frac{3}{5}$	$\frac{3}{5}$	$\frac{252}{5}$
x_3	-1	-2	1	0	-1	0	1	-1	16
x_6	$\frac{4}{5}$	$\frac{9}{10}$	0	0	$\frac{3}{5}$	1	$-\frac{6}{5}$	$\frac{6}{5}$	$\frac{4}{5}$
z^a	0	0	0	0	0	0	0	1	z^a
z^1	$-\frac{2}{5}$	$\frac{4}{5}$	0	0	$\frac{6}{5}$	0	$\frac{3}{5}$	$-\frac{3}{5}$	$\frac{2748}{5}$

↓

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b	
x_4	$\frac{7}{5}$	$\frac{11}{5}$	0	1	$\frac{4}{5}$	0	$-\frac{3}{5}$	$\frac{252}{5}$	$L_1 = L_1 - \frac{7}{5}L_3'$
x_3	-1	-2	1	0	-1	0	1	16	$L_2 = L_2 + L_3'$
→ x_6	$\frac{4}{5}$	$\frac{9}{10}$	0	0	$\frac{3}{5}$	1	$-\frac{6}{5}$	$\frac{4}{5}$	$L_3 = L_3 \cdot \frac{5}{4}$
z^1	$-\frac{2}{5}$	$\frac{4}{5}$	0	0	$\frac{6}{5}$	0	$\frac{3}{5}$	$\frac{2748}{5}$	$L_4 = L_4 + \frac{2}{5}L_3'$

$\min (36, 1)$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_4	0	$\frac{5}{8}$	0	1	$-\frac{1}{4}$	$-\frac{7}{4}$	$\frac{3}{2}$	49
x_3	0	$-\frac{7}{8}$	1	0	$-\frac{1}{4}$	$\frac{5}{4}$	$-\frac{1}{2}$	17
x_1	1	$\frac{9}{8}$	0	0	$\frac{3}{4}$	$\frac{5}{4}$	$-\frac{3}{2}$	1
z'	0	$\frac{5}{4}$	0	0	$\frac{3}{2}$	$\frac{1}{2}$	0	550

$$x^* = (1 \ 0 \ 17 \ 49 \ 0 \ 0 \ 0) \quad z^* = 550$$

d) $\max z = 3x_1 + 4x_2$ $\min z' = -3x_1 - 4x_2$

s.a.: $\begin{cases} 2x_1 + 1x_2 \leq 6 \\ 2x_1 + 3x_2 \leq 9 \\ \vec{x} \in \mathbb{Z}_+^2 \end{cases}$ $\rightarrow$ s.a.: $\begin{cases} 2x_1 + 1x_2 + x_3 = 6 \\ 2x_1 + 3x_2 + x_4 = 9 \\ \vec{x} \in \mathbb{Z}_+^4 \end{cases}$

	x_1	x_2	x_3	x_4	b	
x_3	2	1	1	0	6	$l_1 = l_2 - l_3'$
$\rightarrow x_4$	2	3	0	1	9	$l_2 = l_2 \div 3$
	-3	-4	0	0	0	$l_3 = l_3 + 4l_2'$

$\min(6, 3)$

	x_1	x_2	x_3	x_4	b	
$\rightarrow x_3$	$\frac{4}{3}$	0	1	$-\frac{1}{3}$	3	$l_1 = l_1 \cdot \frac{3}{4}$
x_2	$\frac{2}{3}$	1	0	$\frac{1}{3}$	3	$l_2 = l_2 - \frac{2}{3}l_1'$
	$-\frac{1}{3}$	0	0	$\frac{4}{3}$	12	$l_3 = l_3 + \frac{1}{3}l_1'$

$\min(\frac{9}{4}, \frac{9}{2})$

	x_1	x_2	x_3	x_4	b
x_1	1	0	$\frac{3}{4}$	$-\frac{1}{4}$	$\frac{9}{4}$
x_2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{2}$
	0	0	$\frac{1}{4}$	$\frac{5}{4}$	$\frac{51}{4}$

$$0x_1 - \frac{1}{2}x_3 + \frac{1}{2}x_4 = \frac{3}{2}$$

$$(0+0)x_1 + (-1 + \frac{1}{2})x_3 + (0 + \frac{1}{2}x_4) = (1 + \frac{1}{2})$$

$$\frac{1}{2}x_3 + \frac{1}{2}x_4 \geq \frac{1}{2} \quad (\div \frac{1}{2})$$

$$x_3 + x_4 \geq 1 \rightarrow x_3 + x_4 - x_5 + w_1 = 1$$

$$\min \quad z^a = w_1 = 1 - x_3 - x_4 + x_5$$

	x_1	x_2	x_3	x_4	x_5	w_1	b
x_1	1	0	$\frac{3}{4}$	$-\frac{1}{4}$	0	0	$\frac{9}{4}$
x_2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	$\frac{3}{2}$
w_1	0	0	1	1	-1	1	1
z^a	0	0	0	0	0	1	z^a
z^1	0	0	$\frac{1}{4}$	$\frac{5}{4}$	0	0	$\frac{5}{4}$

$L_4 = L_4 - L_3$

↓

	x_1	x_2	x_3	x_4	x_5	w_1	b	
x_1	1	0	$\frac{3}{4}$	$-\frac{1}{4}$	0	0	$\frac{9}{4}$	$L_1 = L_1 - \frac{3}{4}L_3$
x_2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	$\frac{3}{2}$	$L_2 = L_2 + \frac{1}{2}L_3$
→ w_1	0	0	1	1	-1	1	1	$L_3 = L_3$
z^a	0	0	-1	-1	1	0	z^{a-1}	$L_4 = L_4 + L_3$
z^1	0	0	$\frac{1}{4}$	$\frac{5}{4}$	0	0	$\frac{5}{4}$	$L_5 = L_5 - \frac{1}{4}L_3$

$\min(3, -3, 1)$

	x_1	x_2	x_3	x_4	x_5	w_1	b
x_1	1	0	0	-1	$\frac{3}{4}$	$-\frac{3}{4}$	$\frac{3}{2}$
x_2	0	1	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	2
x_3	0	0	1	1	-1	1	1
z^a	0	0	0	0	0	1	z^a
z^1	0	0	$\frac{1}{4}$	1	$\frac{1}{4}$	0	$\frac{25}{4}$

	x_1	x_2	x_3	x_4	x_5	b
x_1	1	0	0	-1	$\frac{3}{4}$	$\frac{3}{2}$
x_2	0	1	0	1	$-\frac{1}{2}$	2
x_3	0	0	1	1	-1	1
z'	0	0	0	1	$\frac{1}{4}$	$\frac{25}{2}$

$$x_1: -1x_4 + \frac{3}{4}x_5 = \frac{3}{2}$$

$$(-1+0)x_4 + (0+\frac{3}{4})x_5 = (1+\frac{1}{2})$$

$$\frac{3}{4}x_5 \geq \frac{1}{2} \rightarrow \frac{3}{4}x_5 - x_6 + w_2 = \frac{1}{2}$$

	x_1	x_2	x_3	x_4	x_5	x_6	w_2	b	
x_1	1	0	0	-1	$\frac{3}{4}$	0	0	$\frac{3}{2}$	$L_1 = L_1 - \frac{3}{4}L_4'$
x_2	0	1	0	1	$-\frac{1}{2}$	0	0	2	$L_2 = L_2 + \frac{1}{2}L_4'$
x_3	0	0	1	1	-1	0	0	1	$L_3 = L_3 + L_4'$
w_2	0	0	0	0	$\frac{3}{4}$	-1	1	$\frac{1}{2}$	$L_4 = L_4 \cdot \frac{4}{3}$
z''	0	0	0	0	$-\frac{3}{4}$	1	0	$z'' - \frac{1}{2}$	$L_5 = L_5 + \frac{3}{4}L_4'$
z'	0	0	0	1	$\frac{1}{4}$	0	0	$\frac{25}{2}$	$L_6 = L_6 - \frac{1}{4}L_4'$

$\min(2, \frac{2}{3})$

	x_1	x_2	x_3	x_4	x_5	x_6	w_2	b
x_1	1	0	0	-1	0	1	-1	1
x_2	0	1	0	1	0	$-\frac{2}{3}$	$\frac{2}{3}$	$\frac{7}{3}$
x_3	0	0	1	1	0	$-\frac{4}{3}$	$\frac{4}{3}$	$\frac{5}{3}$
x_5	0	0	0	0	1	$-\frac{4}{3}$	$\frac{4}{3}$	$\frac{2}{3}$
z''	0	0	0	0	0	0	1	z''
z'	0	0	0	1	0	0	0	$\frac{37}{3}$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_1	1	0	0	-1	0	1	1
x_2	0	1	0	1	0	$-2/3$	$7/3$
x_3	0	0	1	1	0	$-4/3$	$5/3$
x_5	0	0	0	0	1	$-4/3$	$2/3$
z'	0	0	0	1	0	$1/3$	$37/3$

$$x_2: 1x_4 - 2/3x_6 = 7/3$$

$$(1+0)x_4 + (-1 + 1/3)x_6 = (2 + 1/3)$$

$$1/3x_6 \geq 1/3 \rightarrow x_6 - x_7 + w_3 = 1$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_3	b	
x_1	1	0	0	-1	0	1	0	0	1	$L_1 = L_1 - L_5$
x_2	0	1	0	1	0	$-2/3$	0	0	$7/3$	$L_2 = L_2 + 2/3 L_5$
x_3	0	0	1	1	0	$-4/3$	0	0	$5/3$	$L_3 = L_3 + 4/3 L_5$
x_5	0	0	0	0	1	$-4/3$	0	0	$2/3$	$L_4 = L_4 + 4/3 L_5$
$\rightarrow w_3$	0	0	0	0	0	1	-1	1	1	$L_5 = L_5$
z^a	0	0	0	0	0	-1	1	0	$z^a - 1$	$L_6 = L_6 + L_5$
z'	0	0	0	1	0	$1/3$	0	0	$37/3$	$L_7 = L_7 - 1/3 L_5$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	w_3	b
x_1	1	0	0	-1	0	0	1	-1	0
x_2	0	1	0	1	0	0	$-2/3$	$2/3$	3
x_3	0	0	1	1	0	0	$-4/3$	$4/3$	3
x_5	0	0	0	0	1	0	$-4/3$	$4/3$	2
x_6	0	0	0	0	0	1	-1	1	1
z^a	0	0	0	0	0	0	0	1	z^a
z'	0	0	0	1	0	0	$1/3$	$-1/3$	12

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
x_1	1	0	0	-1	0	0	1	0
x_2	0	1	0	1	0	0	$-\frac{2}{3}$	3
x_3	0	0	1	1	0	0	$-\frac{4}{3}$	3
x_5	0	0	0	0	1	0	$-\frac{4}{3}$	2
x_6	0	0	0	0	0	1	-1	1
z'	0	0	0	1	0	0	$\frac{1}{3}$	12

$$x^* = (0 \ 3 \ 3 \ 2 \ 1) // \quad z^* = 12 //$$